ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION 2024-25

B. Tech. (CSE)

Semester: IV

SUBJECT CODE: BCS404

SOFTWARE ENGINEERING

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define software engineering.

(ii) What is the purpose of a feasibility study?

(iii) What does SEI-CMM stand for?

(iv) Define cyclomatic complexity.

(v) What is regression testing?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is reverse engineering?

(ii) What is COCOMO?

(iii) What is software quality assurance (SQA)?

(iv) List any two characteristics of good software.

(v) What does SRS stand for in software engineering?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Explain the phases of the Software Development Life Cycle.

OR

(b) What is the role of feasibility study in software engineering?

(ii)(a) Compare software verification and validation with examples.

OR

(b) Explain coupling and cohesion in software design. How do they impact software quality?

Q.4.(i)(a) Describe the object-oriented design approach with its advantages.

OR

(b) Differentiate between white box testing and black box testing.

(ii)(a) Explain the prototype model with a neat diagram and describe its phases.

OR

(b) What are the objectives of acceptance testing? Who performs it and when?

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Describe the Software Requirement Specification and its components. Why is SRS important in software development?

OR

(b) Explain the importance of software risk management. How is risk analyzed and managed in a software project?

Q.6.(a) Describe static testing techniques such as code reviews, walkthroughs, and inspections.

OR

(b) What is software re-engineering? Explain its process, benefits, and role in legacy system modernization.